Divide and conquer exercises

04-E01 Maximum-subarray states and interval

Let end(i) be the best nonempty contiguous interval sum ending at i. List every end and best value for A=[5,−9,2,6,−1,4,−3] and recover the optimum interval. What changes on all-negative input if an empty interval is allowed?

04-E02 Recursion trees and fixed-constant induction

Let $T_1=1$ and $T(n)=3T(n/3)+n$ for powers of three. Find the exact solution by level costs and verify it with a fixed-constant inductive upper bound. Why does “cn+n is still O(n)” not prove linear time?

04-E03 A logarithm-squared boundary

For $T(n)=9T(n/3)+n^2(\lg n)^2$ with constant-size bases, explain why none of the three classical master-theorem cases applies directly. Derive a tight bound from level sums.

04-E04 Unequal splits and perturbations

Analyze $T(n)=T(n/3)+T(2n/3)+n$ with positive bounded constant-size bases. State sufficient Akra–Bazzi assumptions, existence and uniqueness of p, the integral, and the result. Why can $T(n)=T(n/2+n/4)+T(n/2)+n$ not keep two 1/2 ratios?

04-E05 Seven products and factor order

Using M1 through M7 defined in the notes, compute every product and C for A=[[2,−1],[3,4]],B=[[5,2],[−2,1]]. Prove the C_{22} formula by block algebra and explain why de cannot be rewritten as ed.

04-E06 Maximum-subarray merging without scanning

Return total, best nonempty prefix, best nonempty suffix, and best subarray for each nonempty interval. Give all four combination formulas for adjacent intervals, prove them generally, and derive total time. Why does this not contradict the CROSS version's Θ(n log n) cost?

04-E07 Strengthening induction to pay lower-order costs

For powers of two, let $T_1=1,T(n)=5T(n/2)+n^2$ and $p=\lg 5$. Where does $T(n)\le cn^p$ fail to close? Use cn^p-dn^2 instead, find feasible c,d, and prove $\Theta(n^p)$.

Divide and conquer exercises (continued)

04-E08 Rounding and base cases

Does T(n)=T(ceil(5n/6))+1 with only T_1=1 define a value for every positive integer? Give a counterexample, a suitable fixed base interval, and the repaired tight bound. Prove that rounding error stays bounded by a constant.

04-E09 Regularity cannot be omitted

For n=2^L let f(n)=n^2 when L is even and n^3 when L is odd. With T_1=1,T(n)=2T(n/2)+f(n), prove that f=Ω(n^2) does not imply T=Θ(f). Identify the failed hypothesis.

04-E10 Finer boundaries than polynomial separation

Find tight bounds for 2T(n/2)+n/lg n and 2T(n/2)+n/(lg n)^2, using positive constant bases for n≤2 and initially restricting n to powers of two. Explicitly give the sums reindexed from the leaves.

04-E11 Parameter copying changes time

Binary search operates on an array of original length N. Compare O(1) index views, copying the current length-n interval on every call, and copying all original N elements regardless of current interval. Give worst-case recurrences, solutions, and cost justifications.

04-E12 Restore the original variable

For $n = 2^{2^k}, k \geq 0$, let T_2=1,T(n)=3T(√n)+lg n. Find a tight bound, identifying the new variable and base case and restoring n. Which variable is confused by the answer Θ(n^(lg 3))?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study